File: `src/components/LiveReadout.jsx`

Category: React Component (Live Value Table)
Runtime: Browser (React)
Primary parent: `src/components/ShellyDashboard.jsx`
Primary data source: latest `ParsedDataPoint` (from `useShellyStatus`)
Upstream dependencies: `src/config/seriesCatalog.js` (metric definitions + grouping)

Overview

`LiveReadout` displays the **latest** device sample as a grouped table of metrics (temperature, current, voltage, etc.). It also shows:

- the time of the last update (clock)
- how long ago the last update occurred ("age")
- a "WARN / STALE" status if the data is old

This component is read-only: it does not modify global state.

Imports

- React: `useEffect`, `useMemo`, `useState`
- `SERIES_CATALOG`, `ALL_METRIC_KEYS`, `getGroupedMetricKeys` from `../config/seriesCatalog`
- `cardStyle` from `../styles/uiStyles`

Internal Helper Functions

`formatAge(ms)`

Signature

```
formatAge(ms: number) → string
```

Purpose

Converts an age in milliseconds to a readable string.

Behavior

- invalid or negative → `"–"`
- under 10 seconds → `"Xs"` with 1 decimal (e.g., `2.4s`)
- under 60 seconds → rounded seconds (e.g., `42s`)
- under 60 minutes → `"Mm SSs"` (e.g., `3m 07s`)
- 60+ minutes → `"Hh Mm"` (e.g., `1h 12m`)

`getAgeStatus(ageMs)`

Signature

```
getAgeStatus(ageMs: number) → { label: string, color: string }
```

Purpose
Maps "age" to a status indicator.

Thresholds

- 15 seconds → `{ label: "STALE", color: "#ff6b6b" }`
- 10 seconds → `{ label: "WARN", color: "#ffd43b" }`
- otherwise → `{ label: "", color: "#ffffffaa" }`

`formatValue(v, decimals)`

Signature

```
formatValue(v: unknown, decimals: number | undefined) → string
```

Purpose
Formats metric values for display.

Behavior

- `null` / `undefined` / `NaN` → `"N/A"`
- number with decimals → fixed decimals
- otherwise → `String(v)`

`formatClock(ts)`

Signature

```
formatClock(ts: Date) → string
```

Purpose

Formats a timestamp as `HH:mm:ss` (24-hour) using `en-GB` locale.

Exported Component

`LiveReadout({ data })`

Signature

```
LiveReadout(props: { data: ParsedDataPoint | null }) → JSX.Element
```

Parameters

- `data`:
  - latest parsed sample (`ParsedDataPoint`)
  - may be `null` while loading

Returns

- A loading message if `data` is missing
- Otherwise, a styled table of metric values + timestamp info

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Internal State and Effects

Local state: `forceTick`

```
const [, forceTick] = useState(0)
```

Purpose

Used only to force periodic re-rendering so “age” updates smoothly even when `data` does not change.

Timer effect

Structure

```
useEffect(() => { setInterval(...) }, [])
```

Dependency array explanation

- Empty dependency array means:
  - timer is created once when the component mounts
  - timer is cleared when component unmounts

Behavior

- Every 200 ms, increment `forceTick`
- This causes the component to re-render and recompute the "age" text

Side effects

- Creates a repeating timer (`setInterval`)
- Cleans up timer on unmount

Derived (Memoized) Values

`grouped`

```
useMemo(() => getGroupedMetricKeys(ALL_METRIC_KEYS), [])
```

Purpose
Precomputes the grouped metric list once.

Dependency array explanation

- Empty array because the metric catalog is static in this application.
- Grouping does not depend on `data`.

`lastMs` and `ageMs`

- `lastMs` comes from `data.timestamp.getTime()`
- `ageMs = Date.now() - lastMs`

These are recomputed each render; the timer-driven re-render ensures `ageMs` changes over time.

`ageText` and `ageStatus`

Computed with `useMemo` based on `ageMs`.

This avoids re-running formatting/status logic unnecessarily during the same render.

Rendering Logic

Loading guard

If `data` is null:

- renders `Loading Device data...`

Main table

For each group and metric key:

1. Look up metric in `SERIES_CATALOG`
2. If it has a `valueAccessor`, call it:
   - `value = valueAccessor(data)`
3. Render row:
   - label
   - formatted value
   - unit

Last update footer

Shows:

- clock time only when status is normal (not WARN/STALE)
- always shows "(age ago • WARN/STALE)" when applicable

Side effects

- Timer for re-rendering ("age" updates)
- No network activity
- No global state changes